

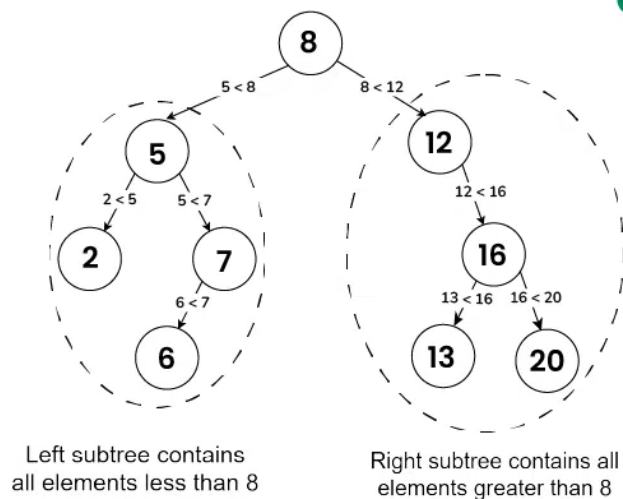
Introduction to Binary Search Tree

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Binary Search Tree is a data structure used in computer science for organizing and storing data in a sorted manner. Binary search tree follows all properties of binary tree and for every nodes, its **left** subtree contains values less than the node and the **right** subtree contains values greater than the node. This hierarchical structure allows for efficient **Searching**, **Insertion**, and **Deletion** operations on the data stored in the tree.

Binary Search Tree



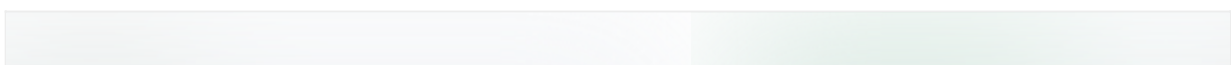
Binary Search Tree

Properties of Binary Search Tree:

- The left subtree of a node contains only nodes with keys less than the node's key.
- The right subtree of a node contains only nodes with keys greater than the node's key.
- The left and right subtree each must also be a binary search tree.
- There must be no duplicate nodes(BST may have duplicate values with different handling approaches).

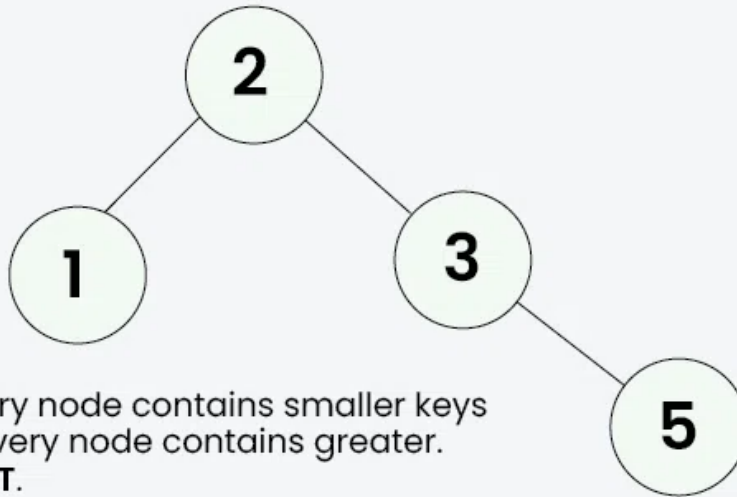
Test Your Understanding

Given a Binary Tree, find out if it is Binary Search Tree or not.



Example 01

Output: True



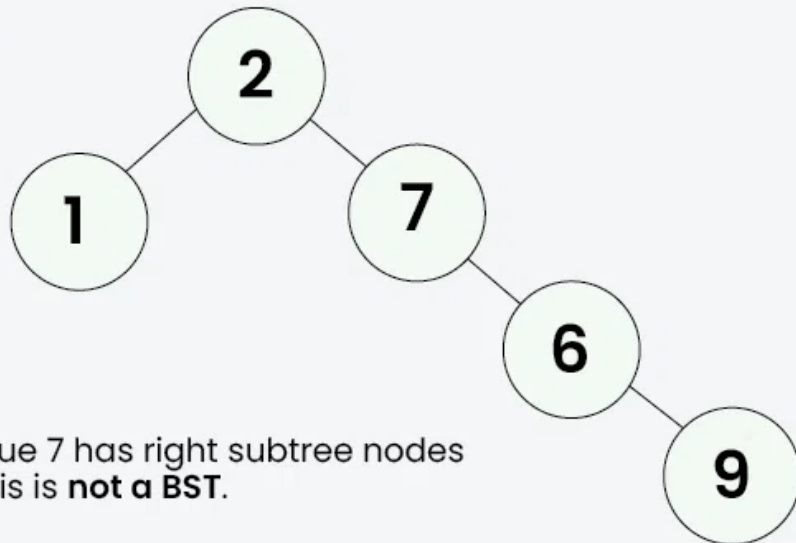
Explanation:

The left subtree of every node contains smaller keys and right subtree of every node contains greater. Hence, the tree is a **BST**.

Check if a Binary Tree is BST or not

Example 02

Output: False



Explanation:

Since the node with value 7 has right subtree nodes with keys less than 7, this is **not a BST**.

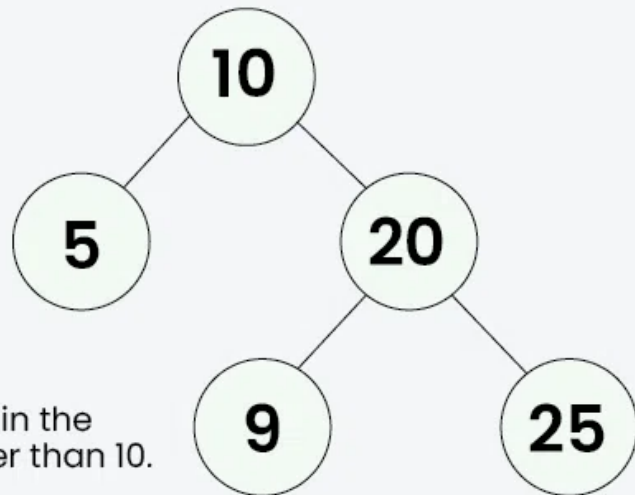
Check if a Binary Tree is BST or not

Example 03

Output: False

Explanation:

The node with value 9 is present in the right subtree of 10, but it is smaller than 10. Hence, the tree is **not a BST**.



————— Check if a Binary Tree is BST or not —————

Important Points about BST

- A Binary Search Tree is useful for maintaining sorted stream of data. It allows search, insert, delete, ceiling, max and min in $O(h)$ time. Along with these, we can always traverse the tree items in sorted order.
- With Self Balancing BSTs, we can ensure that the height of the BST is bound by $\log n$. Hence we achieve, the above mentioned $O(h)$ operations in $O(\log n)$ time.
- When we need only search, insert and delete and do not need other operations, we prefer [Hash Table](#) over BST as a Hash Table supports these operations in $O(1)$ time on average.

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Insertion in Binary Search Tree (BST)

Given a BST, the task is to insert a new node in this BST.Example: How to Insert a value in a Binary Search Tree:A new key is always inserted at the leaf by maintaining the property of the binary search...

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Searching in Binary Search Tree (BST)

Given a BST, the task is to search a node in this BST. For searching a value in BST, consider it as a sorted array. Now we can easily perform search operation in BST using Binary Search Algorithm. Input: Root of...

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Deletion in Binary Search Tree (BST)

Given a BST, the task is to delete a node in this BST, which can be broken down into 3 scenarios:Case 1. Delete a Leaf Node in BSTDeletion in BSTCase 2. Delete a Node with Single Child in BSTDeleting a...

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Given a Binary Search Tree, The task is to print the elements in inorder, preorder, and postorder traversal of the Binary Search Tree.Â Input:Â A Binary Search TreeOutput:Â Inorder Traversal: 10 20 30 100 150...

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Balance a Binary Search Tree

Given a BST (Binary Search Tree) that may be unbalanced, the task is to convert it into a balanced BST that has the minimum possible height.Examples: Input: Output: Explanation: The above unbalanced BST...

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Self-Balancing Binary Search Trees are height-balanced binary search trees that automatically keep the height as small as possible when insertion and deletion operations are performed on the tree. The heigh...

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